Beyond the Average Treatment Effect

Modified treatment policies

MODIFIED

April 23, 2025

WEBR STATUS

Ready!

As we already discussed, a modified treatment policy is an intervention characterized by a dependence on the natural, or observed, value of treatment. Examples include:

- Decreasing surgeries that last more than 4 hours by 1 hour (additive MTP)
- Decreasing BMI by 2 units for individuals with BMI > 30 (additive MTP)
- Decreasing AQI by 20% (multiplicative MTP).

Synthetic Neighborhood Community Violence Data

In some current work, we are interested in estimating the effect of decreasing the level of community violence (violent crime rate) on risk of preterm birth. Research suggests community violence may impact infant health through physiological responses to stress, worsening maternal mental health, and unhealthy coping behaviors. Community violence is a highly endogenous exposure, which, motivates the use of such a shift intervention.[1]

owner_vacancy_rate	renter_vacancy_rate	poverty_line_prop	unemployed_prop	no_car_prop	never_married_prop	separated_prop	mater
0.7	2.6	9.1	33.4	36.1	44.5	5.1	24.0
6.9	8.8	41.6	11.5	40.5	34.2	5.1	37.0
1.5	16.6	17.3	6.2	9.6	53.8	2.2	24.0
11.4	1.1	34.1	11.2	18.7	37.3	6.1	34.0
6.3	2.4	26.5	18.8	51.4	25.0	3.0	22.0
1.5	5.4	4.4	16.5	8.0	56.6	2.1	41.0
14.2	0.7	10.8	22.5	15.5	32.5	3.2	40.0
0.0	9.4	6.2	34.6	8.6	23.7	4.6	48.0
8.3	5.7	23.6	12.7	19.7	51.3	1.4	23.0
4.3	0.0	24.1	8.0	20.9	66.5	3.5	44.0

Previous 1 2 3 4 5 ... 100 Next

Problem 1

Write a shift function (named `d_mtp`) for a modified treatment policy where the natural value of neighborhood violence is decreased by 50%. Using the `lmtp` package, estimate the population mean effect of this intervention on preterm birth. Make sure to use the `id` argument to account for community level clustering in the estimation of standard errors. To save computation time, don't use cross-fitting.

Run Code

```
A <- "neighborhood_violence"
W <- names(crime)[1:12]
Y <- "preterm_birth"
id <- "community_code"

learners <- c("SL.mean",
              "SL.glm",
              "SL.rpart",
              "SL.gam")

d <- # Write your shift function here!

fit_d <- # Write your estimation code here!
```

▶ ✅ Solution

Did you get this error?

```
Warning: Detected decimalish `trt` values and `mtp = FALSE`. Consider setting `mtp =
TRUE` if getting errors.
Loading required package: nnls
Error: object 'fit' not found
```

> Make sure to set `mtp = TRUE` if you're intervention is a modified treatment policy!

Problem 2

Let's compare the effect of the intervention on reducing neighborhood violence on preterm birth to the observed preterm birth rate. Use the `lmtp_contrast()` function.

Run Code

```
# Write your code here!
```

▶ ✅ Solution

With `lmtp` we found that under a hypothetical intervention that reduces neighborhood violence by 50%, preterm birth would decrease by approximately 4.4 percentange points.

References

Díaz, Iván, Nicholas Williams, Katherine L Hoffman, and Edward J Schenck. 2023. "Nonparametric Causal Effects Based on Longitudinal Modified Treatment Policies." *Journal of the American Statistical Association* 118 (542): 846–57.

Haneuse, Sebastian, and Andrea Rotnitzky. 2013. "Estimation of the Effect of Interventions That Modify the Received Treatment." *Statistics in Medicine* 32 (30): 5260–77.

Footnotes

1. Thank you to Chris Dharma, who simulated this data! ↩